

Day 18 - Lecture 18.17: Pie Charts (plt.pie)

1. Overview & Mathematical Definition

A **Pie Chart** is a circular graphic divided into slices to illustrate numerical proportions of a whole (100%).

- Each slice represents a category.
- The arc length, central angle, and area of each slice are proportional to the quantity it represents.

Central Angle Formulation:

$$\theta_i = 360^\circ \times \frac{y_i}{\sum_{j=1}^K y_j}$$

Percentage of Total:

$$p_i = 100\% \times \frac{y_i}{\sum_{j=1}^K y_j}$$

2. Syntax & Parameters

```
plt.pie(  
    x,                # Slice magnitudes (array of values)  
    labels=None,      # Sequence of strings providing labels for each slice  
    autopct='%1.1f%%', # String or function used to format slice value percentage  
    colors=None,      # List of color codes for slices  
    startangle=0,     # Counter-clockwise rotation angle from the x-axis  
    explode=None,     # Fractional radius offset to pull slices outwards  
    shadow=False      # Draw a 3D drop shadow beneath pie  
)
```

3. Code Implementation: Corporate Expense Breakdown

```
import matplotlib.pyplot as plt

expenses = ["Salaries", "Rent", "Marketing", "R&D", "Miscellaneous"]
amounts = [500, 150, 120, 100, 50] # Total = $920K

colors = ["#4C72B0", "#55A868", "#C44E52", "#8172B2", "#CCB974"]

plt.figure(figsize=(7, 7))
plt.pie(
    amounts,
    labels=expenses,
    autopct='%1.1f%%',      # Displays percentage formatted to 1 decimal place
    colors=colors,
    startangle=140
)

plt.title("Company Budget Allocation by Expense Category", fontsize=14, fontweight='bold')
plt.show()
```

4. Key Takeaways & Limitations

- **When to Use:** Part-to-whole relationships with **few categories (3 to 5 max)**.
- **When NOT to Use:** When comparing more than 6 categories, or comparing two categories of similar sizes (e.g. 24% vs 26%). The human brain is poor at estimating angles and areas compared to estimating lengths in bar charts.